Первое задание

Т1) $\xi \sim R(0;\theta)$; $\theta \in \Theta = (0;+\infty)$

вер-ая модель

выборка $\vec{X}_n$

$\tilde{\theta}_1 = 2\bar{x}$, $\tilde{\theta}_2 = X_{min}$; $\tilde{\theta}_3 = X_{max}$; $\tilde{\theta}_4 = \left(X_1 + \frac{\sum_{k=2}^{n} x_k}{n-1}\right)$

$$M\xi = \int_{-\infty}^{\infty} x\, dF(x,\theta) = \int_0^{\theta} x \cdot \frac{1}{\theta}\, dx = \frac{\theta^2}{2} \cdot \frac{1}{\theta} = \frac{\theta}{2}$$

$$\left\{ \begin{array}{l} p(x,\theta) = \frac{1}{\theta}\{(0;\theta)\} \\ \xi \sim R(a;b);\ p(x) = \frac{1}{b-a}\{(a;b)\} \end{array} \right\}$$

$$M\xi^2 = \int_0^{\theta} x^2 \cdot \frac{1}{\theta}\, dx = \frac{\theta^2}{3}$$

$$D\xi = M\xi^2 - (M\xi)^2 = \frac{\theta^2}{3} - \frac{\theta^2}{4} = \frac{\theta^2}{12}$$

парам. модель

1) $\tilde{\theta}_1$ $\forall \theta > 0$;

$$M\tilde{\theta}_1 = M\left(2 \cdot \frac{1}{n} \sum x_i\right)$$

x_i — незав. случ. вел. : $x_i \sim R(0;\theta)$

$$M\tilde{\theta}_1 = \frac{2}{n} \sum M x_i = \frac{2}{n} \cdot n \cdot M\xi = 2 \cdot \frac{\theta}{2} = \theta \Rightarrow \underline{\text{несмещ.}}$$

$$D\tilde{\theta}_1 = D\left(\frac{2}{n} \sum x_i\right) = \frac{4}{n^2} \sum D x_i = \frac{4}{n^2} \cdot n\, D\xi =$$

$$= \frac{4}{n} \cdot \frac{\theta^2}{12} \xrightarrow[n\to\infty]{} 0 \quad \forall \theta > 0 \Rightarrow \underline{\text{состоят.}}$$ (по дост. усл.)

2) $\tilde{\theta}_2 = X_{min}$

$\forall \theta > 0$ $\quad M\tilde{\theta}_2 = M X_{min}$

$\xi \sim F(x)$

$$\xi_{min} \sim \underbrace{1-(1-F(x))^n}_{\varphi(x)}$$

$$\varphi(x) = \varphi'(x) = n(1-F(x))^{n-1} F'(x) = n\left(1-\frac{x}{\theta}\right)^{n-1} \frac{1}{\theta}\{(0;\theta)\}$$

$$M x_{min} = \int_0^\theta x\, n\underbrace{\left(1-\frac{x}{\theta}\right)}_{t}^{n-1} \frac{1}{\theta} dx = -\int_1^0 n(1-t)t^{n-1}\theta\, dt =$$

$$t;\ x = \theta(1-t)$$

$$= n\theta \int_0^1 (t^{n-1} - t^n) dt = n\theta\left(\frac{1}{n} - \frac{1}{n+1}\right) = \frac{\theta}{n+1} \text{ — смещен.}$$

$$\tilde{\theta}_2' = (n+1) x_{min}$$

$$M\tilde{\theta}_2' = (n+1) M x_{min} = \theta \text{ — несмещ.}$$

$$D\tilde{\theta}_2' = D((n+1)x_{min}) = (n+1)^2 D x_{min}$$

$$M x_{min}^2 = \int_0^\theta x^2 n\underbrace{\left(1-\frac{x}{\theta}\right)}_{t}^{n-1} \cdot \frac{1}{\theta} dx = \int_0^1 \theta^2 (1-t)^2 n t^{n-1} \cdot \frac{\theta}{\theta} dt =$$

$$= n\theta^2 \int_0^1 (t^{n-1} - 2t^n + t^{n+1}) dt = n\theta^2\left(\frac{1}{n} - 2\cdot\frac{1}{n+1} + \frac{1}{n+2}\right) =$$

$$= \frac{2\theta^2}{(n+1)(n+2)}$$

$$(n+1)(n+2) - 2(n+2)n + n(n+1) = \cancel{n^2} + \cancel{3n} + 2 - \cancel{2n^2}$$

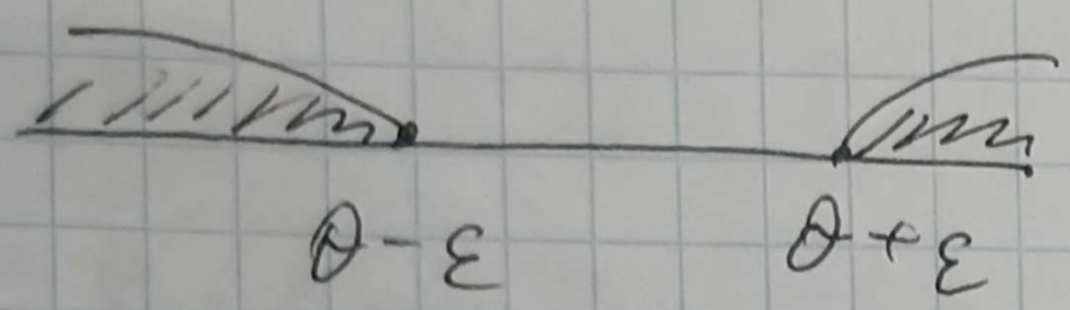
$$- \cancel{2n} + \cancel{n^2} + \cancel{n} = 2$$

$$D x_{min} = \frac{2\theta^2}{(n+1)(n+2)} - \frac{\theta^2}{(n+1)^2} = \theta^2\left(\frac{2(n+1)-(n+2)}{(n+1)^2(n+2)}\right) =$$

$$= \frac{\theta^2 n}{(n+1)^2(n+2)} \xrightarrow[n\to\infty]{} 0$$ — ничего не следует (дост. усл. не работает, т.к. $\tilde{\theta}_2$ смещ)

$$D((n+1)^2 x_{min}) = \frac{\theta^2 n}{n+2} \xrightarrow[n\to\infty]{} 0$$

<u>no</u> ~~obj.~~ : $\forall \theta > 0 \quad \forall \varepsilon > 0 \quad P(|\tilde{\theta}_2' - \theta| \geq \varepsilon) \xrightarrow[n\to\infty]{} 0$



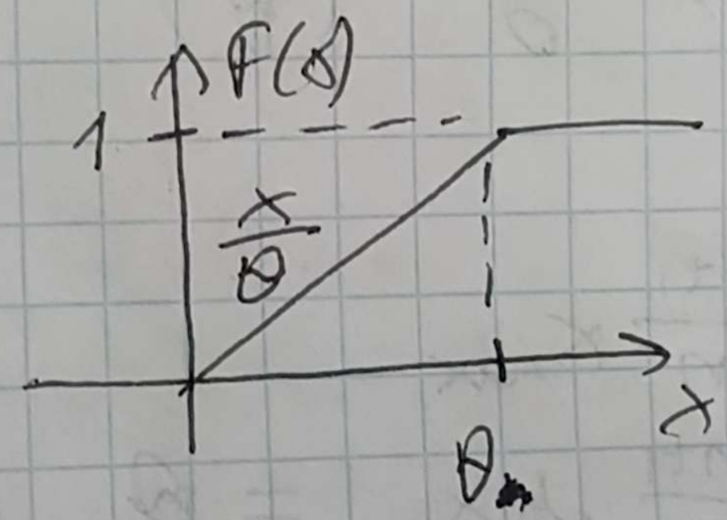
$\tilde{\theta}_2' \geq \varepsilon + \theta$

$\tilde{\theta}_2' \leq -\varepsilon + \theta$

$$P(|\tilde{\theta}_2' - \theta| \geq \varepsilon) \geq P(\tilde{\theta}_2' \geq \theta + \varepsilon) = P((n+1)x_{min} \geq$$

$$\geq \theta + \varepsilon) = P\left(x_{min} \geq \frac{\theta+\varepsilon}{n+1}\right) = 1 - \Phi\left(\frac{\theta+\varepsilon}{n+1}\right) \ominus$$

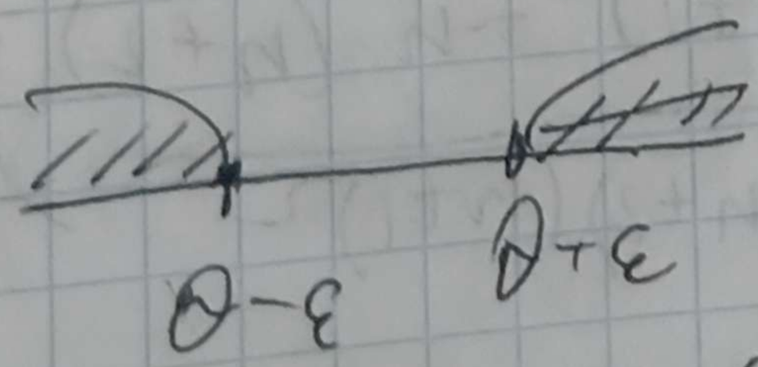
$$\Phi(x) = 1 - (1 - F(x))^n$$



$$\ominus\ 1 - \left(1 - \left(1 - \frac{\theta+\varepsilon}{(n+1)\theta}\right)^n\right) = \left(1 - \frac{\theta+\varepsilon}{(n+1)\theta}\right)^n \xrightarrow[n\to\infty]{}$$

$$\to e^{-\frac{\theta+\varepsilon}{\theta}} > 0 \Rightarrow \underline{\text{не явл. сост.}}$$

$$P(|x_{min} - \theta| \geq \varepsilon) = P(x_{min} \leq \theta - \varepsilon) = \Phi(\theta - \varepsilon) \ominus$$



$P(x_{min} \geq \theta + \varepsilon) = 0$

$$\ominus\ 1 - (1 - F(\theta - \varepsilon))^n = 1 - \left(1 - \frac{\theta - \varepsilon}{\theta}\right)^n =$$

$$= 1 - \left(\frac{\varepsilon}{\theta}\right)^n \xrightarrow[n\to\infty]{} 1$$

$\exists \varepsilon > 0, \ \exists \theta > 0: \quad \frac{\varepsilon}{\theta} < 1$ не явн. сост.

3) $\tilde{\theta}_3 = X_{max}$

$M\tilde{\theta}_3 = M X_{max}$

$X_{max} \sim \underbrace{(F(x))^n}_{\Psi(x)}$

$\varphi(x) = \Psi'(x) = n\left(\frac{x}{\theta}\right)^{n-1} \cdot \frac{1}{\theta} \{(0;\theta)\}$

$M X_{max} = \int_0^\theta x n \left(\frac{x}{\theta}\right)^{n-1} \cdot \frac{1}{\theta} dx = \frac{n}{\theta^n} \frac{\theta^{n+1}}{n+1} = \frac{n}{n+1}\theta$

смещ.

$\tilde{\theta}_3' = \frac{n+1}{n} X_{max}$

$M\tilde{\theta}_3' = \frac{n+1}{n} M X_{max} = \theta$ несмещ

$D\tilde{\theta}_3' = \left(\frac{n+1}{n}\right)^2 D X_{max} \xrightarrow[n\to\infty]{} 0$ состят.

$M X_{max}^2 = \int_0^\theta x^2 n \left(\frac{x}{\theta}\right)^{n-1} \cdot \frac{1}{\theta} dx = \frac{n}{\theta^n} \frac{\theta^{n+2}}{n+2} =$

$= \frac{\theta^2 n}{n+2}$

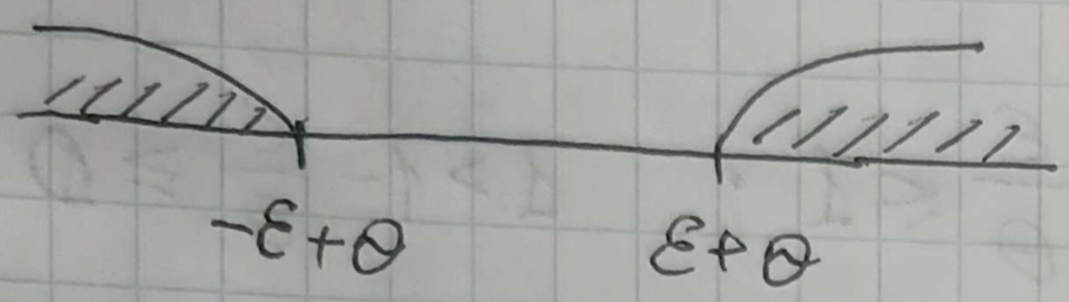
$D X_{max} = \frac{\theta^2 n}{n+2} - \frac{\theta^2 n^2}{(n+1)^2} = \theta^2 \left(\frac{n(n+1)^2 - n^2(n+2)}{(n+2)(n+1)^2}\right) =$

$= \theta^2 \left(\frac{n^3 + 2n^2 + n - n^3 - 2n^2}{(n+2)(n+1)^2}\right) = \theta^2 \frac{n}{(n+2)(n+1)^2}$

по опр.

$\forall \theta>0, \forall \varepsilon>0$

$$\mathbb{P}(|\tilde{\theta}_3' - \theta| \geq \varepsilon) \xrightarrow[n\to\infty]{?} 0 \qquad \cancel{\forall\theta, \forall\varepsilon>0, \forall\theta}$$



$$\mathbb{P}(\tilde{\theta}_3' \geq \varepsilon+\theta) = \mathbb{P}\left(\frac{n+1}{n} x_{max} \geq \varepsilon+\theta\right) =$$

$$= \mathbb{P}\left(x_{max} \geq \frac{n}{n+1}(\varepsilon+\theta)\right)$$

$x_{max} \leq \theta$. $\quad \frac{n}{n+1}(\varepsilon+\theta) < \theta$; $\quad n\varepsilon + n\theta < n\theta + \theta$

$n\varepsilon < \theta$

для фикс. θ и ε $\quad \exists N = \frac{\theta}{\varepsilon}\ \forall n \geq N \to n \geq \frac{\theta}{\varepsilon}$,

т.е. $x_{max} \geq \theta$, т.е. $\mathbb{P}\left(x_{max} \geq \frac{n}{n+1}(\varepsilon+\theta)\right) = 0$

$$\Rightarrow \mathbb{P}(\tilde{\theta}_3' \geq \varepsilon+\theta) \xrightarrow[n\to\infty]{} 0$$

$$\mathbb{P}(\tilde{\theta}_3' \leq -\varepsilon+\theta) = \mathbb{P}\left(\frac{n+1}{n} x_{max} \leq -\varepsilon+\theta\right) =$$

$$= \mathbb{P}\left(x_{max} \leq \frac{n}{n+1}(\theta-\varepsilon)\right) = \left(F\left(\left(\frac{n}{n+1}\right)(\theta-\varepsilon)\right)\right)^n =$$

$$= \left(\frac{n(\theta-\varepsilon)}{(n+1)\theta}\right)^n = \left(\frac{n}{n+1}\left(1-\frac{\varepsilon}{\theta}\right)\right)^n = \left(1-\frac{1}{n+1}\right)^n\left(1-\frac{\varepsilon}{\theta}\right)^n$$

$$\left(1-\frac{1}{n+1}\right)^n = \frac{\left(1+\frac{-1}{n+1}\right)^{n+1}}{\left(1-\frac{1}{n+1}\right)} \xrightarrow[n\to\infty]{} \frac{e^{-1}}{1} = e^{-1}$$

б) Если $\theta-\varepsilon<0$, т.е. $\theta<\varepsilon$, то $\mathbb{P}\left(x_{max} \leq \underbrace{\frac{n}{n+1}(\theta-\varepsilon)}_{<0}\right) = 0$

Поэтому рассмотрим $\theta \geq \varepsilon$:

$\cancel{\left(1-\frac{\varepsilon}{\theta}\right)^n \quad 0<1-\frac{\varepsilon}{\theta}<1}$ –

$0<\varepsilon\le\theta; \quad 0<\frac{\varepsilon}{\theta}\le 1; \quad 1>1-\frac{\varepsilon}{\theta}\ge 0$ (4)

$\Rightarrow \left(1-\frac{\varepsilon}{\theta}\right)^n \xrightarrow[n\to\infty]{} 0$

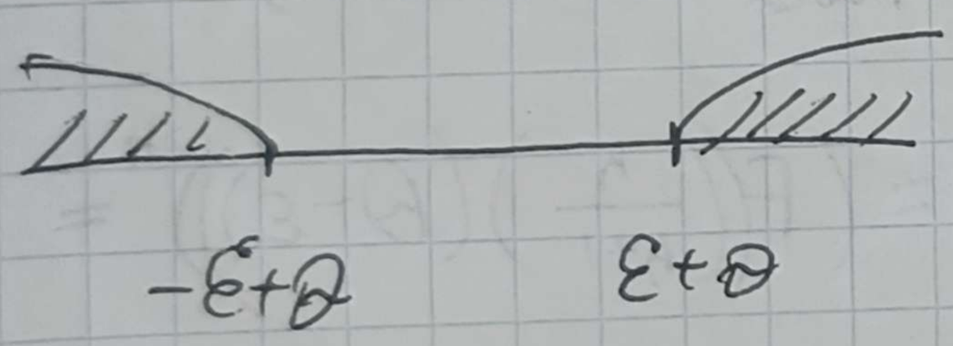
$P\left(X_{max}\le\frac{n}{n+1}(\theta-\varepsilon)\right)\xrightarrow[n\to\infty]{} 0$ (произв. огр. и б.м.)

$\Rightarrow P\left(|\tilde\theta_3'-\theta|\ge\varepsilon\right)\xrightarrow[n\to\infty]{} 0+0=0 \Rightarrow$

$\Rightarrow \tilde\theta_3'$ <u>состоятельная</u>.

для $\tilde\theta_3$ дост. усл. сост. не выполняется (т.к. $\tilde\theta_3$ <u>смещ.</u>). по <u>опр.</u>

$\forall\theta>0, \forall\varepsilon>0 \quad P\left(|\tilde\theta_3-\theta|\ge\varepsilon\right)\overset{?}{\xrightarrow[n\to\infty]{}}0$



$P(X_{max}\ge\varepsilon+\theta)=0$

$P(X_{max}\le\theta-\varepsilon)=\left(F(\theta-\varepsilon)\right)^n=\left(\frac{\theta-\varepsilon}{\theta}\right)^n=$

$=\left(1-\frac{\varepsilon}{\theta}\right)^n$, аналогично как и для $\tilde\theta_3'$

рассмотрим $\varepsilon\le\theta$: $\left(1-\frac{\varepsilon}{\theta}\right)^n\xrightarrow[n\to\infty]{}0$

$\Rightarrow P\left(|\tilde\theta_3-\theta|\ge\varepsilon\right)\xrightarrow[n\to\infty]{}0 \Rightarrow \tilde\theta_3$ <u>состоятельна</u>

④ $\tilde{\theta}_4 = x_1 + \frac{1}{n-1}\sum_{k=2}^{n} x_k$

$M\tilde{\theta}_4 = Mx_1 + \frac{1}{n-1}\sum_{k=2}^{n} Mx_k = M\xi + \frac{1}{n-1}\sum_{k=2}^{n} M\xi =$

$= \frac{\theta}{2} + \frac{\theta}{2} = \theta$ <u>несмещ.</u>

$D\tilde{\theta}_4 = Dx_1 + \frac{1}{(n-1)^2}\sum_{k=2}^{n} Dx_k = \frac{\theta^2}{12} + \frac{1}{(n-1)}\cdot\frac{\theta^2}{12} \Rightarrow$

$= \cancel{\frac{\theta^2}{12}} \xrightarrow[n\to\infty]{} 0$ не выт. дост. усл.

$\tilde{\theta}_4 \xrightarrow{P} \theta$

$x_1 + \frac{1}{n-1}\sum_{k=2}^{n} x_k$

$\left\{ \begin{array}{l} \xi_n \xrightarrow{P} \xi \\ \eta_n \xrightarrow{P} \eta \end{array} \right. \rightarrow \left. \begin{array}{l} \xi_n + \eta_n \xrightarrow{P} \xi + \eta \\ \xi_n \eta_n \xrightarrow{P} \xi\eta \end{array} \right\}$

$x_1 \xrightarrow{P} \xi \qquad \frac{1}{n-1}\sum_{k=2}^{\infty} x_k \xrightarrow{P} Mx_1 = \frac{\theta}{2}$

$\left\{ \begin{array}{l} \underline{\text{ЗБЧ Хинчина}} \quad \xi_n \text{ — независ., одинаково распр.} \\ \exists M\xi_n < \infty \rightarrow \frac{1}{n}\sum_{k=1}^{n} \xi_k \xrightarrow{P} M\xi_1 \end{array} \right\}$

$x_1 + \frac{1}{n-1}\sum_{k=2}^{n} x_n \xrightarrow{P} \xi + \frac{\theta}{2}$ <u>не явл. сост.</u>

б) $\tilde{\theta}_3' = \frac{n+1}{n} x_{max}$ $\qquad D\tilde{\theta}_3' = \frac{\theta^2}{(n+2)n}$

$\tilde{\theta}_1 = 2\bar{x}$ $\qquad D\tilde{\theta}_1 = \frac{\theta^2}{n\cdot 3}$

$$\frac{1}{3n} > \frac{1}{n(n+2)}$$

$$n^2 + 2n > 3n \Rightarrow \tilde{\theta}_3' \text{ эффективнее } \tilde{\theta}_1$$

$$n+2 > 3$$

$$n > 1$$